

Session 3

Putting it all together

Exercise 1

Write out the steps (handout 1)

Instructions

- Identify a sequence of data verbs to transform the Starting Table into the Ending Table
- Use data verbs + some pseudo code
 - Example: `MUTATE(ROUND(Distance))`
- Data Verbs
 - JOIN
 - UNSTACK/PIVOT
 - STACK/UNPIVOT
 - MUTATE
 - RECODE
 - SPLIT STRING
 - IF/THEN
 - GROUPBY + AGGREGATE

Starting Tables

ID	F1	F5	F2	F3.5	F2.11
1	Somewhat Agree	Somewhat Disagree	Somewhat Agree	Somewhat Agree	Somewhat Agree
2	Somewhat Agree	Somewhat Disagree	Somewhat Agree	Neither Agree nor Disagree	Somewhat Agree
3	Strongly Agree	Neither Agree nor Disagree	Somewhat Agree	Strongly Disagree	Somewhat Agree
4	Somewhat Agree	Somewhat Agree	Strongly Agree	Somewhat Disagree	Somewhat Agree
5	Strongly Agree	Strongly Disagree	Neither Agree nor Disagree	Somewhat Disagree	Somewhat Agree

Abc	#	#	Abc	Abc
Question	Construct	Question # on Qua...	Needs Reverse Co...	Column Name
In the future, I plan to ...	1	1	No	F1
Individuals are respon...	5	2	Yes	F5
When trying to unders...	2	3	No	F2
I plan to become invol...	1	4	No	F1.1
I can communicate we...	2	5	No	F2.1

Step	Data Description
1	
2	
3	
4	
5	

Ending Table

Participant	F1	F2	F3	F4	F5	F6
213	40	57	19	15	34	25
69	40	60	23	24	29	21
5	37	47	22	19	36	19
123	29	43	17	19	25	15

Write out the steps (handout 2)

- Identify a sequence of data verbs to transform the Starting Table into the Ending Table
- Use data verbs + some pseudo code
 - Example: `MUTATE(ROUND(Distance))`
- Data Verbs
 - `SELECT`
 - `FILTER`
 - `MUTATE`
 - `GROUPBY + AGGREGATE`
 - `PARTITIONBY + ORDERBY`

Table				Data Description
ID	F1	F5	F2	NA
1	Somewhat Agree	Somewhat Disagree	Somewhat Agree	
2	Somewhat Agree	Somewhat Disagree	Somewhat Agree	
3	Strongly Agree	Neither Agree nor Disagree	Somewhat Agree	
4	Somewhat Agree	Somewhat Agree	Strongly Agree	
5	Strongly Agree	Strongly Disagree	Neither Agree nor Disagree	

Note: Starting with the health survey data			
Participant	Question	Response	
1	F1	Somewhat Agree	
1	F1.1	Somewhat Agree	
1	F1.2	Somewhat Agree	
1	F1.3	Somewhat Agree	
1	F1.4	Neither Agree nor Disagree	
1	F1.5	Somewhat Agree	
1	F1.6	Somewhat Agree	

Note: One row per question					
Participant	Question	Response	Question (Question ID)	Construct	Question # on Qualtr...
1	F1	Somewhat Agree	In the future, I plan to participate in a community service or...	1	1
1	F1.1	Somewhat Agree	I plan to become involved in my community	1	4
1	F1.2	Somewhat Agree	I plan to help others who are in difficulty	1	10
1	F1.3	Somewhat Agree	I plan to do some volunteer work	1	19
1	F1.4	Neither Agree nor Disagree	I plan to become involved in programs to help clean up the e...	1	28

Note: Combine health survey and recoding tables					
Participant	Question	Response	Needs Revers...	TempRecode	TempReverse
36	F4.1	Strongly Disagree	Yes	1	5
57	F4.1	Strongly Disagree	Yes	1	5
61	F4.1	Strongly Disagree	Yes	1	5
62	F4.1	Strongly Disagree	Yes	1	5

Note: Recoding values: Regular coding – Strongly Disagree = 1 ... Strongly Agree = 5; Reverse coding – Strongly Disagree = 5 ... Strongly Agree = 1

Participant	Question	Response	Needs Revers...	TempRecode	TempReverse	Recorded Val...
36	F4.1	Strongly Disagree	Yes	1	5	5
57	F4.1	Strongly Disagree	Yes	1	5	5
61	F4.1	Strongly Disagree	Yes	1	5	5
62	F4.1	Strongly Disagree	Yes	1	5	5

Hint: Use IF							
Participant	Question	Response	Needs Revers...	TempRecode	TempReverse	Recorded Val...	Question Type
36	F4.1	Strongly Disagree	Yes	1	5	5	F4
57	F4.1	Strongly Disagree	Yes	1	5	5	F4
61	F4.1	Strongly Disagree	Yes	1	5	5	F4
62	F4.1	Strongly Disagree	Yes	1	5	5	F4

Note: Create Question Type variable		
Participant	Question Type	Recorded Val...
130	F6	21
254	F1	17
130	F4	24
71	F4	21
241	F6	18

Note: Find total score per question type per participant						
Participant	F1	F2	F3	F4	F5	F6
181	39	57	25	23	32	20
235	37	47	23	17	29	15
62	34	53	23	23	31	19
19	36	55	21	19	39	21
3105	31	44	25	26	38	14

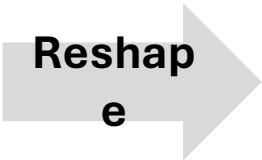
Note: Split questions into columns

Response Table

ID	F1	F5
1	Somewhat Agree	Somewhat Disagree
2	Somewhat Agree	Somewhat Disagree
3	Strongly Agree	Neither Agree nor ...
4	Somewhat Agree	Somewhat Agree
5	Strongly Agree	Strongly Disagree

Agree?

Disagree?



Stacked Response Table

ID	Question	Response
1	F1	Somewhat Agree
1	F5	Somewhat Disagree
1	F2	Somewhat Agree
1	F1.1	Somewhat Agree
1	F2.1	Somewhat Agree



Needs Reverse Coding?	Column Name
No	F1
Yes	F5
No	F2
No	F1.1
No	F2.1
Yes	F6

Question Table

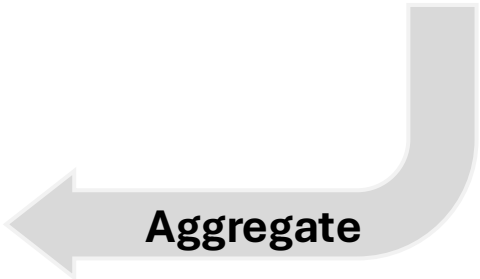


Joined Table

ID	Question	Response	Needs Reverse Coding?
1	F1	Somewhat Agree	No
1	F5	Somewhat Disagree	Yes
1	F2	Somewhat Agree	No
1	F1.1	Somewhat Agree	No
1	F2.1	Somewhat Agree	No



Question	Type	Response	Regular Coding	Reverse Coding	Coding
F1	F1	Somewhat Agree	4	2	2
F5	F5	Somewhat Disagree	2	4	2
F2	F2	Somewhat Agree	4	2	2
F1.1	F1	Somewhat Agree	4	2	2
F2.1	F2	Somewhat Agree	4	2	2



ID	F1	F2	F3	F4	F5	F6
1	17	24	16	13	20	12
2	17	25	17	13	21	10
3	12	20	17	12	16	13
4	16	18	24	15	18	14
5	11	25	14	11	12	11

ID	Type	Total
1	F1	17
1	F2	24
1	F3	16
1	F4	13
1	F5	20